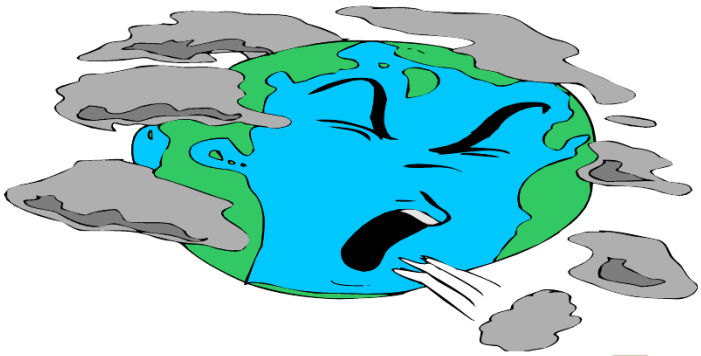
Air Pollution

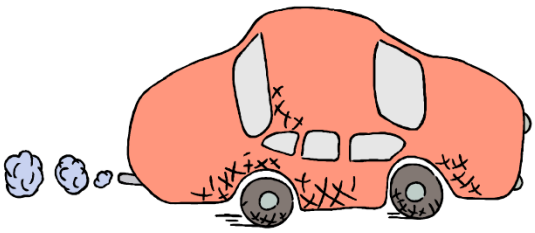
Any visible or invisible particle or gas found in the air that is not part of the original, normal composition.



Air Pollution

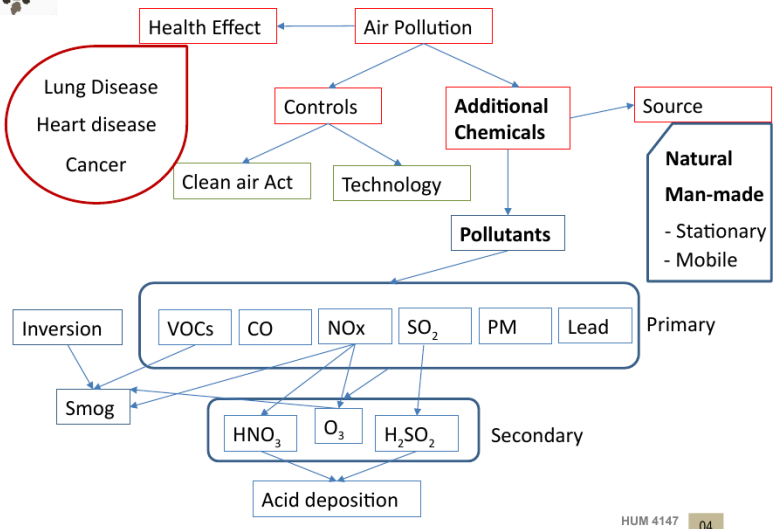
Natural: forest fires, pollen, dust storm

Unnatural: man-made; coal, wood and other fuels used in cars, homes, and factories for energy



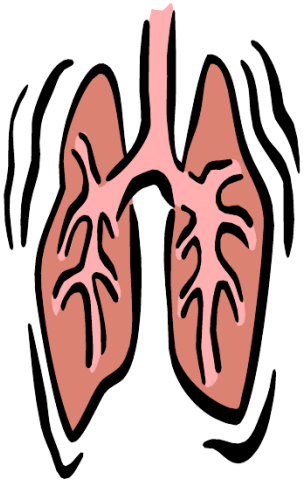


Air Pollution



Source: Bozeman Science

Air Pollution



Nitrogen Dioxide

- reddish, brown gas
- produced when nitric oxide combines with oxygen in the atmosphere
- present in car exhaust and power plants
- affects lungs and causes wheezing; increases chance of respiratory infection

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Air Pollution

Particulate Matter

- particles of different sizes and structures that are released into the atmosphere
- present in many sources including fossil fuels, dust, smoke, fog, etc.
- can build up in respiratory system
- aggravates heart and lung disease; increases risk of respiratory infection



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Air Pollution

Ground Level Ozone

- at upper level, ozone shields Earth from sun's harmful UV rays
- at ground level, ozone is harmful pollutants
- formed from car, power and chemical plant exhaust
- irritate respiratory system and asthma; reduces lung function by inflaming and damaging lining of lungs



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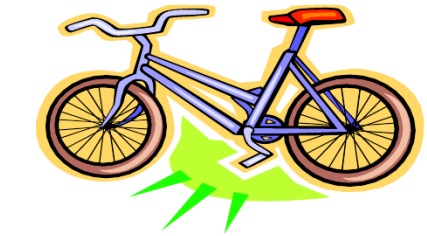
Air Pollution

Air Quality Index Levels of Health Concern	Numerical Value	Meaning
Good	0-50	Air quality is considered satisfactory, and air pollution poses little or no risk.
Moderate	51-100	Air quality is acceptable; however, for some pollutants there may be a moderate health concern for a very small number of people who are unusually sensitive to air pollution.
Unhealthy for Sensitive Groups	101-150	Members of sensitive groups may experience health effects. The general public is not likely to be affected.
Unhealthy	151-200	Everyone may begin to experience health effects; members of sensitive groups may experience more serious health effects.
Very Unhealthy	201-300	Health alert: everyone may experience more serious health effects.
Hazardous	> 300	Health warnings of emergency conditions. The entire population is more likely to be affected.

Air Pollution

What can we do?

- Ride your bike
- Tell your friends and family about pollution
- Make sure your parents get pollution checks on their cars
- Ride the school bus





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Air Pollution

- Learn more; stay up to date
- Join a group to stop pollution
- Encourage your parents to carpool to work
- Switch off lights, fan etc. when you leave the room/house.



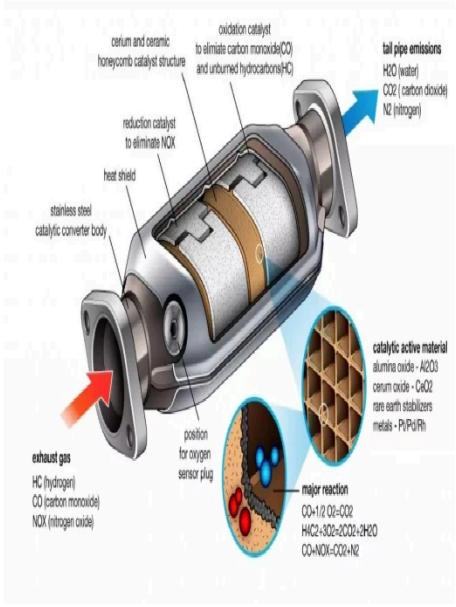


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4. by using electret filter material (usually, electrospun plastic fibers) to attract or repel particles with an electrostatic charge, so that they are more likely to collide with the filter surface
5. by using certain coatings on the fibers that kill or deactivate infectious particles colliding with them (such as salt)
6. by using gravity and allowing particles to settle into the filter material (this effect is typically negligible); and
7. by using the particles themselves, after the filter has been used, to act as a filter medium for other particles.

Catalytic converter

A catalytic converter is a unit that fits into the front part of exhaust system of an engine, **to reduce the emission of gaseous pollutants, such as carbon monoxide (CO), nitrogen oxides (NOx) and hydrocarbons (HCs)**.its an exhaust emission control device that converts toxic gases and pollutants in exhaust gas from an internal combustion engine into less-toxic pollutants by catalyzing a redox reaction.



Catalytic converters are usually used with internal combustion engines fueled by gasoline or diesel, including lean-burn engines, and sometimes on kerosene heaters and stoves. The catalyst formulation comprises three key constituents: precious metals (platinum, palladium, rhodium), alumina, and rare-earth-based materials which enhance catalytic activity of the metals. These "two-way" converters combine oxygen with carbon monoxide (CO) and unburned hydrocarbons (HC) to produce carbon dioxide (CO_2) and water (H_2O). Although catalytic converters are most commonly applied to exhaust systems in automobiles, they are also used on electrical generators, forklifts, mining equipment, trucks, buses, locomotives, motorcycles, and on ships. They are even used on some wood stoves to control emissions. This is usually in response to government regulation, either through environmental regulation or through health and safety regulations.